A rank collapse in a sedenion Module-LWE proposal, and why a uniform matrix does not repair it

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October 2026 (revised)

Abstract

We report a negative result on SLWE, a Module-LWE variant over the 16-dimensional sedenion algebra that the author proposed in an earlier paper with a claim of NIST Category 5 security. Its public matrix is built from orbits of an order-7 coordinate permutation (a "Singer cycle"). First, the flattened public matrix factors through a fixed 256×112 integer matrix of rank 76, so its rank over $\mathbb{F}_q$ is at most 76 for every module rank k, and exactly 76 for generic seeds once $k \geq 7$, whereas a uniform matrix of the same shape has rank $16k$. This invalidates the security claim at both published parameter sets. Recovering the noise then reduces to an LWE instance of dimension 76, which the lattice estimator places at about 2^{39} operations at the first set ($n = 512$, $q \approx 2^{32}$) and 2^{40} at the second ($n = 1024$, $q = 911$). Second, replacing the matrix by a uniform one does not restore the claim (centred binomial noise with $\eta = 2$, ternary secrets of weight 64): the estimator gives about 2^{51} at the first set and about 2^{167} at the second, against roughly 2^{272} for Category 5. Third, decryption at the first set never fails: the noise is bounded by 258, far below $q/4$. All computations are reproducible from public scripts.

1 Introduction

SLWE was proposed inside a larger, physics-motivated programme that uses the sedenions $\mathbb{S}$ (the Cayley–Dickson double of the octonions) and their zero divisors. The idea was to keep the shape of Module-LWE [14, 10] while replacing the ring by $\mathbb{S}_q = \mathbb{S} \otimes \mathbb{F}_q$ and to give the public matrix a symmetry inherited from the Fano plane: each row is the orbit of a random seed under a cyclic permutation σ of order 7. Correctness was obtained with a bilinear "norm" pairing rather than the (non-associative) algebra product.

Both versions of the paper that proposed SLWE [5] claimed NIST Category 5 security. Version 1 used $n = 512$ and $q \approx 2^{32}$. Version 2 found that modulus far too large (its primal estimate put version 1 below Category 1), moved to $k = 64$ and $q = 911$ ($n = 1024$), and claimed Category 5 from a primal estimate, leaving the dual and hybrid attacks to be checked; Section 4 checks them. It kept the Singer-orbit matrix, whose rank it had tested only at $k = 4$. The project's working specification [6] recorded on May 10, 2026 that this matrix is rank-deficient at $k = 32$, and replaced it by a uniform one.

This note shows that neither version is secure and records three facts that may be useful beyond this proposal.

1. A coordinate permutation applied to seeds produces a matrix that is linear in the seeds through a fixed "design matrix". The rank of that design matrix bounds the rank of the public matrix for every module rank (Section 3).

2. Non-associativity of the algebra plays no role once decryption uses a bilinear form: every map the attacker needs is $\mathbb{F}_q$-linear on $\mathbb{F}_q^{16k}$.

3. At $q \approx 2^{32}$ with noise of standard deviation about 1, LWE in dimension 512 is easy whatever the matrix (Section 4). The modulus-to-noise ratio, not the algebra, governs security. Version 2 of [5] reached the same conclusion with a primal estimate; Section 4 confirms it with the standard estimator.

2 The scheme

Let $\mathbb{S}_q$ be the sedenions over $\mathbb{F}_q$ with basis $e_0, \dots, e_{15}$, built by four Cayley–Dickson doublings $(a,b)(c,d) = (ac - \bar{d}b,\ da + b\bar{c})$. Write $L(a) \in \mathbb{F}_q^{16\times 16}$ for left multiplication by a and $\langle u, v\rangle = \mathrm{Re}(\bar{u}v) = \sum_t u_t v_t$ for the norm pairing, extended to $\mathbb{S}_q^k$ by summing over components. The identity $L(a)^{\mathsf{T}} = L(\bar{a})$ gives $\langle As, r\rangle = \langle s, A^H r\rangle$ with $A^H_{ij} = \overline{A_{ji}}$, which is what decryption needs.

- **KeyGen.** $s \in \mathbb{S}_q^k$ is sparse ternary (weight h_s). $A \in \mathbb{S}_q^{k\times k}$ has $A_{ij} = \sigma^j(\mathrm{seed}_i)$ with independent uniform seeds, where σ moves the coefficient of e_t to $e_{(t \bmod 7)+1}$ for $t = 1..7$, moves that of e_{t+8} to $e_{(t \bmod 7)+9}$, and fixes e_0, e_8. The noise e is centred binomial (or confined to the four-dimensional kernel of a zero divisor [12]). The public key is $(A,\ b = As + e)$.
- **Encaps.** r is sparse ternary (weight h_r), with $c_1 = A^H r + e_1$ and $c_2 = \langle b, r\rangle + e_2 + m\lfloor q/2\rceil$.[1]
- **Decaps.** Round $c_2 - \langle s, c_1\rangle = m\lfloor q/2\rceil + N$ with $N = \langle e, r\rangle - \langle s, e_1\rangle + e_2$.

Version 1 specifies $n = 512$ (so $k = 32$), a modulus $q \approx 2^{32}$ with $q \equiv 1 \pmod{455}$, and sparse ternary secrets. We instantiate it with the least prime above 2^{32} that is $\equiv 1 \pmod{455}$, $q = 4\,294\,977\,961$, and with $h_s = h_r = 64$ and CBD($\eta = 2$) noise, the values of version 2 and of the project's ledger [9]. Version 2 specifies $k = 64$ ($n = 1024$), $q = 911$ (the least prime $\equiv 1 \pmod{455}$), $h_s = h_r = 64$ and CBD(2) noise.

3 The rank collapse

Let $M \in \mathbb{F}_q^{16k\times 16k}$ be the matrix of $s \mapsto As$, i.e. $M_{16i+u,\,16j+c} =$ coefficient u of $\sigma^j(\mathrm{seed}_i)\, e_c$.

Proposition 1. *Let $N \in \mathbb{Z}^{256\times 112}$ have entries $N_{16m+u,\,16j+c} =$ coefficient u of $\sigma^j(e_m)\, e_c$ ($0 \le m, u, c \le 15$, $0 \le j \le 6$). Then $M = (S \otimes I_{16})\, N_k$, where $S \in \mathbb{F}_q^{k\times 16}$ holds the seeds and N_k consists of the column blocks $j \bmod 7$ of N, $j = 0, \dots, k-1$. Hence* $\mathrm{rank}\, M \le \mathrm{rank}\, N$ *for every k. Moreover* $\mathrm{rank}_{\mathbb{Q}}\, N = 76$, *and* $\mathrm{rank}\, N = 76$ *modulo* 911 *and modulo q.*

Proof. The product is bilinear and σ is linear, so $\sigma^j(\mathrm{seed}_i)\, e_c = \sum_m (\mathrm{seed}_i)_m\, \sigma^j(e_m)\, e_c$. That is the stated factorisation. Since $\sigma^7 = \mathrm{id}$, column block j of N_k equals block $j \bmod 7$, so N_k has at most the columns of N. The ranks of N were computed exactly (Section 7). □

The bound is attained. Table 1 lists exact ranks from two independent implementations (one built on the proposal's reference code, one written blind from a textual specification), with three seed sets per case and a uniform control. The collapse starts at $k = 5$; from $k = 7$ on the rank is 76 for every seed set tried. Version 2's parameters, $k = 64$ over $\mathbb{F}_{911}$, are in the table: the rank is 76, not 1024. Its own rank test used $k = 4$, where the matrix has 64 columns, below the cap of 76, so the collapse could not show there.

Remark 1. σ is not an automorphism of the sedenion product, so the "symmetry" of A is a relabelling of coordinates. The argument above uses only linearity.

[1] For odd q, $\lfloor q/2\rceil$ stands for $\lfloor q/2\rfloor$ or $\lceil q/2\rceil$; the reference code uses $\lfloor q/2\rfloor$. Section 5 treats both.

k	1	2	3	4	5	6	7	8	16	32	64
$16k$	16	32	48	64	80	96	112	128	256	512	1024
rank M (Singer)	16	32	48	64	72	74	76	76	76	76	76
rank M (uniform)	16	32	48	64	80	96	112	128	256	512	1024

Table 1: Exact ranks over $\mathbb{F}_{911}$ and over $\mathbb{F}_q$ with $q = 4\,294\,977\,961$, identical at both primes for three seed sets each, in both implementations.

Corollary 1. *Let V be the column space of M ($\dim V = \rho \le 76$). Then $e \in b + V$. Choose ρ coordinates I on which a basis of V is invertible. Then $e_{\bar{I}} = -Ce_I + (b_{\bar{I}} + Cb_I)$ for a public C, an LWE instance with secret e_I of dimension ρ and $16k - \rho$ samples. The same holds for e_1 and the column space of M^H. Given e and e_1, any r' with $M^H r' = c_1 - e_1$ gives $c_2 - \langle b - e, r' \rangle = m\lfloor q/2 \rceil + \langle e, r \rangle + e_2$, which reveals m.*

For $\rho = 76$, $q = 4\,294\,977\,961$, CBD(2) secret and noise and $m = 436$ samples, the lattice estimator [2, 3] reports $2^{38.6}$ operations (uSVP and BDD). Its block size is at the floor of its model ($\beta = 40$), so reduction at the scale of LLL already suffices. Distinguishing (A, b) from uniform is easier still: it needs only Gaussian elimination, because A itself is rank-deficient. At version 2's parameters the same reduction gives dimension 76 with 948 samples, and the estimator reports $2^{40.4}$ (BDD hybrid, again at $\beta = 40$).

4 A uniform matrix does not restore the claimed security

After the rank deficiency was found, the project replaced the Singer matrix by a uniform one. Brief 06 [7] adds a uniform perturbation to it, which at the strength chosen there ($\delta = 1$) makes the matrix uniform, and puts it into the key generation of the project's hybrid-KEM wrapper, whose target is $k = 32$ with $q \approx 2^{32}$ [8]. The working specification [6] states that its security table applies with this matrix. Table 2 shows the estimator's default attack set with a uniform matrix at version 1's setting as instantiated in Section 2, at the two SLWE rows of the security table of [6], at $q = 2731$, and at version 2's parameters. At version 1's setting the cheapest attack costs about 2^{51}. An independent implementation of the primal "2016 estimate" [1] (smallest β with $\sigma\sqrt{\beta} \le \delta_\beta^{2\beta-d-1}\mathrm{Vol}^{1/d}$, the secret rescaled to the noise width) gives $\beta = 72$, $d = 937$, a core-SVP cost of $2^{0.292\beta} = 2^{21}$ [4], the same as the estimator's own rough uSVP figure. The cause is the modulus: with $q/\sigma \approx 2^{32}$, a block size near 70 already exposes the short vector in dimension about 940. A dense ternary secret instead of weight 64 changes little: about 2^{53}.

n	q	claimed	cheapest attack ($\log_2$ rop)	uSVP ($\log_2$ rop, β)
512	4 294 977 961	Cat. 5 (v. 1)	51.0 (BDD)	52.1, $\beta = 72$
512	3329	L3 [6]	112.2 (dual hybrid)	120.3, $\beta = 322$
512	2731	—	114.4 (dual hybrid)	122.5, $\beta = 330$
512	911	L5 [6]	122.2 (BKW)[a]	136.9, $\beta = 383$
1024	911	Cat. 5 (v. 2)	167.2 (MITM hybrid)[b]	253.9, $\beta = 800$

Table 2: Lattice estimator [3], commit `53da598`, default attacks and cost model [11], with a uniform public matrix, $m = n$ samples, ternary secrets of weight 64 and CBD(2) noise. "Claimed" is NIST Category 5 for the rows of versions 1 and 2 of [5], and the level label of the security table of [6] for its rows. Version 1's row uses the instantiation of Section 2; 2731 is the second-smallest prime $\equiv 1 \pmod{455}$. [a]BKW needs about $2^{110.7}$ samples, and a public key gives 512; the cheapest other attack costs $2^{127.4}$. [b]Without its meet-in-the-middle step the hybrid costs $2^{181.1}$.

The working specification [6] targets $n = 512$ with "∼3329 (mod-455 prime)". Its security table labels $q = 911$ "L5" (408 bits classical) and $q = 3329$ "L3" (341 bits), from a formula calibrated on Kyber that assumes a full-rank matrix. With weight-64 secrets neither row reaches 2^{128}: $q = 911$ gives at most about 2^{127} and $q = 3329$ about 2^{112}. A dense ternary secret lifts $q = 911$ only to about 2^{149}. The ML-KEM modulus 3329 is not $\equiv 1 \pmod{455}$; the nearest prime below it that is, 2731, gives about 2^{114}. At $n = 512$, then, even the smallest admissible prime gives at most about 2^{127} with weight-64 secrets and CBD(2) noise.

Version 2 took a larger dimension. With a uniform matrix, the estimator's cheapest attack on its parameters is the primal hybrid with a meet-in-the-middle step, at about 2^{167}; without that step the hybrid costs 2^{181}. The other lattice attacks cost 2^{239} (dual hybrid), 2^{249} (BDD), 2^{254} (primal uSVP, the attack version 2 used, at $\beta = 800$ against its claimed $\beta \approx 942$) and 2^{268} (dual). NIST's thresholds are about 2^{143} classical gates for Category 1, 2^{207} for Category 3 and 2^{272} for Category 5 [13]; version 2 measured its own gate count against the last. Reading the estimator's operation counts as gate counts, every lattice attack in the default set is below 2^{272}, so version 2's Category 5 claim fails even with a uniform matrix, and the primal hybrid attacks put its parameters below Category 3. We have not searched the space of q, η and secret weight further.

5 Correctness

Each coordinate of e, e_1 and e_2 lies in $[-2, 2]$. The confined zero-divisor noise has the same bound, since the kernel bases have entries in $\{0, \pm 1\}$ and the noise is $2\sum_j \alpha_j k_j$ with $\alpha_j \in \{-1, 0, 1\}$. So

$$|N| \le 2h_r + 2h_s + 2 = 258 < \lfloor q/4 \rfloor \approx 1.07 \times 10^9,$$

and decryption never fails at version 1's setting. The bound $258 < q/4$ alone would suggest the same for every $q > 1032$, but at the boundary the half unit lost in rounding $q/2$ matters: at $q = 1033$, a prime, the worst case $|N| = 258$ decodes wrongly with either rounding of $q/2$ and under every decoding rule we tried (thresholds at $q/4$ and $3q/4$, the nearest point, rounding $2v/q$, and the reference code's branchless rounding). From $q = 1036$ on it never fails under any of them, so the first prime at which decryption never fails is 1039. At version 2's $q = 911$ it can fail, though rarely. For CBD noise the distribution is in fact exact: given r and s, N is a sum of $2(h_r + h_s + 1)$ independent symmetric binomial draws, i.e. $N \sim \mathrm{CBD}(258)$. At $q = 911$ the tail is $\Pr[|N| > 227] = 2^{-353.5}$. The failure rate itself, for a uniform message bit, depends on the rounding at the boundary: it is $2^{-353.9}$ for every rule and rounding above except the reference code's decoder with its encoding $m\lfloor q/2 \rfloor$, which gives $2^{-351.4}$. Both are consistent with version 2's Gaussian bound of 2^{-295}.

The project's working ledger [9] reported $\log_2 \mathrm{DFR} \approx -6.4 \times 10^{15}$ for version 1's setting. That number is the Gaussian tail at $z = q/(4\sigma_N) \approx 9.5 \times 10^7$ with $\sigma_N \approx 11.4$. It evaluates the tail about 4×10^6 times beyond the largest value N can take. At that modulus the correct statement is $\mathrm{DFR} = 0$.

6 Conclusion

The Singer-orbit public matrix makes SLWE insecure at every module rank, including both published parameter sets. A uniform matrix does not restore the claimed security: version 1's setting reaches about 2^{51}, and version 2's parameters about 2^{167}, below Category 5 under every lattice attack in the estimator's default set. The scheme's correctness was never in question. The general lesson is elementary but easy to miss. A structured matrix built by acting on random seeds is the image of a fixed design matrix, and that matrix's rank should be the first thing checked.

7 Reproducibility

All scripts, outputs and checksums are in the public repository `gifgaf0/gifgaf0.github.io` at commit

`d96d1f7451ffca96468ce1918f595370154558af`

in the directory `audit_followup/phase_c/C1/`:

- the two rank implementations: `leg1/`, and `leg2/`, written blind from the pre-registration `C1_PREREG.md`;
- their comparison and the computation of rank N: `c1_compare.py`;
- the estimator runs: `estimator/`, using passagemath 10.8.12 and lattice-estimator commit `53da5982597709ba0fdf94ea37a84d822310fd84`;
- the core-SVP cross-check: `c1_coresvp_crosscheck.py`;
- the noise bounds and exact tails: `c1_checks.py`.

The runs added in this revision are in the directory

`audit_followup/closeout/step3/3b_slwe_eprint/v2_runs/`:

- Table 1 for $k \leq 7$, with three seed sets and a uniform control at both primes in both implementations: `table1_k1to7.py`;
- the estimator at version 2's parameters, at the $q = 3329$ row of [6] and with dense ternary secrets: `v2_estimator_run.py`;
- the exact failure rates and the decoder scan of Section 5: `dfr_exact.py`.

Exact ranks were computed with python-flint.

Changes in this revision. The first version of this note (doi:10.5281/zenodo.23270249) contained four errors, corrected here. Its abstract cited a failure rate of $2^{-6.4\times10^{15}}$ as "reported earlier", which reads as a claim of [5]; the figure comes from the project's ledger [9]. It said that [6] kept version 1's parameters; that document targets $q \approx 3329$. It presented the prime $4\,294\,977\,961$ and the weight 64 as version 1's own; they are our instantiation. And it said that decryption never fails for any $q > 1032$, which is false at $q = 1033$. It also described Table 1 as having three seed sets and a uniform control for every k, which held only for $k \geq 8$; the runs for $k \leq 7$ are now made, with the same values. This revision also gives the failure rate at $q = 911$ next to the tail, adds estimates for version 2's parameters and for the $q = 3329$ row of [6], and compares both rows of [6] with their claims.

Acknowledgements. AI agents (Anthropic's Claude) drafted this note and wrote and ran all of its computations, including both independent rank implementations. A separate Claude agent audited the first version of this note and found the errors corrected here. The author directed the work, reviewed the text, and takes responsibility for it.

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